

Connor N. Monaghan

Available For Co-op in Spring and Summer Semester of 2027

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Education

University of Cincinnati, Cincinnati, OH
Bachelor of Science - *Computer Engineering*

Graduating: April 2028
GPA: 3.76

University of Cincinnati, Cincinnati, OH
Master of Engineering - *Electrical Engineering*

Graduating: April 2028
GPA: N/A

Experience

R.A Jones, Covington, KY

Electrical Engineering Co-op

May 2026 – August 2026

- Developed a high-speed inspection application to detect 1mm defects within a 500ms window using a Keyence 2D/3D Laser Profiler
- Facilitated the deployment and commissioning of a next generation vision system at an overseas customer facility
- Collected and presented actionable data from real production to reduce downtime and minimize material waste
- Partnered with executive leadership, customer stakeholders, and Keyence field engineers to ensure a highly accurate, custom vision application
- Designed and wired a custom industrial control panel integrating an Allen-Bradley PLC, 3 servo drives, 3 VFDs, multiple relays and distributed I/O
- Engineered the foundational software for a Beckhoff XTS system to control 80 independent movers on a state-of-the-art packaging machine prior to commissioning

August 2025 – December 2025

- Integrated an Allen-Bradley PLC and a Keyence vision system to automatically detect length, width, height and rotation of tissue stacks at a rate of 320 per minute, automatically rejecting bad stacks
- Programmed dynamic part-tracking logic in Studio 5000 to synchronize reject timing across 7 independent, variable-speed conveyors prior to automated solenoid rejection
- Led development and programming of a Keyence 2D/3D Laser Profiler to interface with an Allen-Bradley PLC on next generation packaging equipment

January 2025 – April 2025

- Implemented custom Allen-Bradley ladder logic during Factory Acceptance Testing (FAT) to resolve a key client requirement, ensuring machine approval and sign off
- Enhanced existing core machine logic and an Allen-Bradley PanelView HMI to improve efficiency and reduce programming time for future models

Personal Projects

Home Lab – Self Hosted Infrastructure

- Repurposed a 2012 laptop to host a portfolio website, a Kali Linux virtual machine and two separate Minecraft servers
- Configured local resource allocation and network routing to maintain reliable performance across multiple concurrent services

Web Development – Personal Website

- Developed and hosted a client-side web application to showcase personal projects, technical skills, and accomplishments

Home Assistant – Smart Home Controller

- Deployed a centralized Home Assistant interface to monitor and control personal devices and custom applications such as lighting, 3D prints and commercially available devices

Relevant Skills

- **Industrial Automation:** Rockwell Automation (Studio 5000 Ladder logic, ControlLogix, CompactLogix), Beckhoff (TwinCAT 3 Structured Text, XTS) Keyence Measurement sensors, IO-Link
- **Languages:** Python, C++, LabView, Java, Visual Basic, MATLAB
- **Developer Tools:** Git, Linux, Portainer, Docker, Excel